

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
7	(a)			A disease which is always present at low levels (in an area)/ frequently at a predictable rate in a specific location (1)	1			1		
7	(b)	(i)		A – disulphide bridge/disulphide bond B – Antigen binding site C – Light chain D – Heavy chain All 4 correct = 3 marks, 3 correct = 2 marks, 2 correct =1 mark	3			3		
		(ii)		Antigen –initiates an {immune response/production of antibodies} (1) Antigen-antibody complex- a <u>specific</u> antibody bound to an antigen/ complementary antibody (1)	2			2		
	(c)	(i)		<u>Memory cells</u> (must make IgG during the secondary response/ after second exposure/clonal expansion)(1)		1		1		
		(ii)		Any 2 x (1) from: - The bacteria may have antigenic variation/surface proteins / reference to mutations (1) (IgM produced again as) no memory cells to new antigen (1) - Primary response each time (1)			2	2		
	(d)	(i)		Delay in detection of bacterial antigen /reference to latent period/clonal selection (1) IgG not produced straight away / IgG non existent until day 7/ Time is needed to produce and secrete antibodies (1)		2		2		

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		(ii)		Any 5 x (1) from: A. Antibodies maybe already attached to the bacteria/antigen/ a substance within the plasma (1) B. Different strain of bacteria which is not recognised by the test (1) C. more than one type of antigen on the surface of the bacteria (1) D. antibodies not at detectable levels/ Variation amongst peoples' immune response/ colour change not detectable (1) E. Enzyme not functioning/denatured (1) F. lack of control of variables e.g. pH/ temperature/ volume of sample (1)		2	3	5		
	(e)	(i)		35-38 °C and it is a human pathogen/ similar to body temperature (1)		1		1		1
		(ii)		380/ 380.13/ 379.94 mm ² = 2 marks Allow the following for 1 mark 380 (no units) (22/2) ² x 3.14 = (11) ² v 3.14 =		2		2	2	
		(iii)		The zone of inhibition is a perfect circle/ radius is the same all the way round (1)		1		1		
				Question 7 total	6	9	5	20	2	3